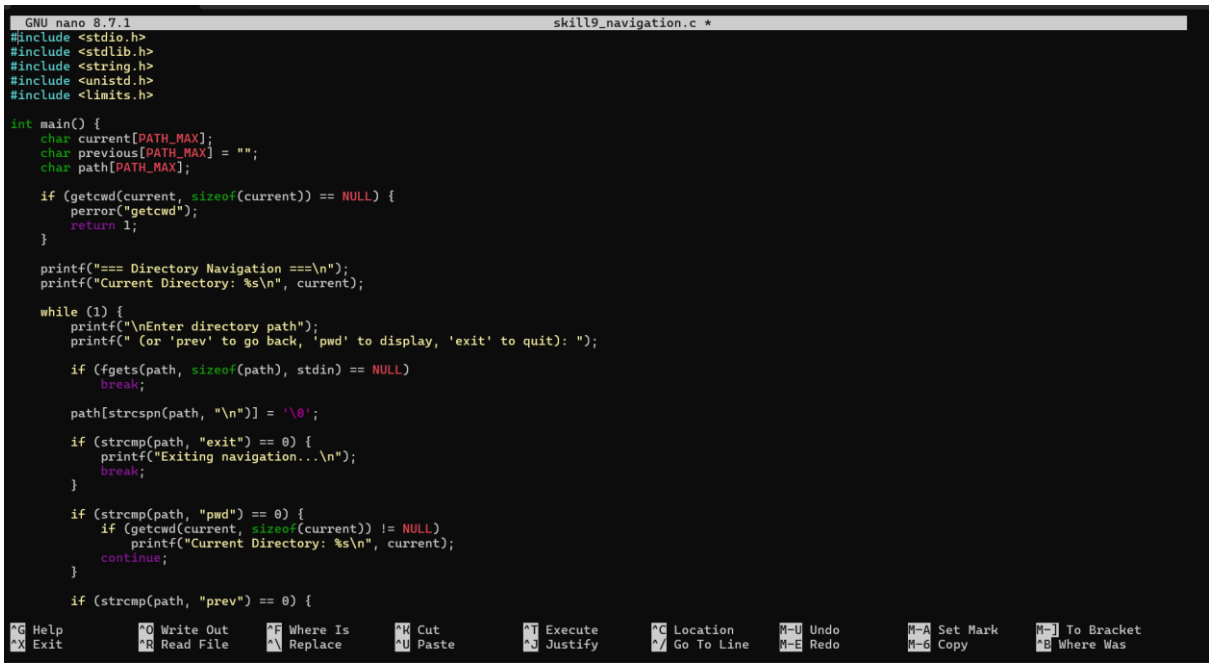
Skill-09.c



```
 GNU nano 8.7.1                                  skill9_navigation.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>

int main() {
    char current[PATH_MAX];
    char previous[PATH_MAX] = "";
    char path[PATH_MAX];

    if (getcwd(current, sizeof(current)) == NULL) {
        perror("getcwd");
        return 1;
    }

    printf("=== Directory Navigation ===\n");
    printf("Current Directory: %s\n", current);

    while (1) {
        printf("\nEnter directory path");
        printf(" (or 'prev' to go back, 'pwd' to display, 'exit' to quit): ");

        if (fgets(path, sizeof(path), stdin) == NULL)
            break;

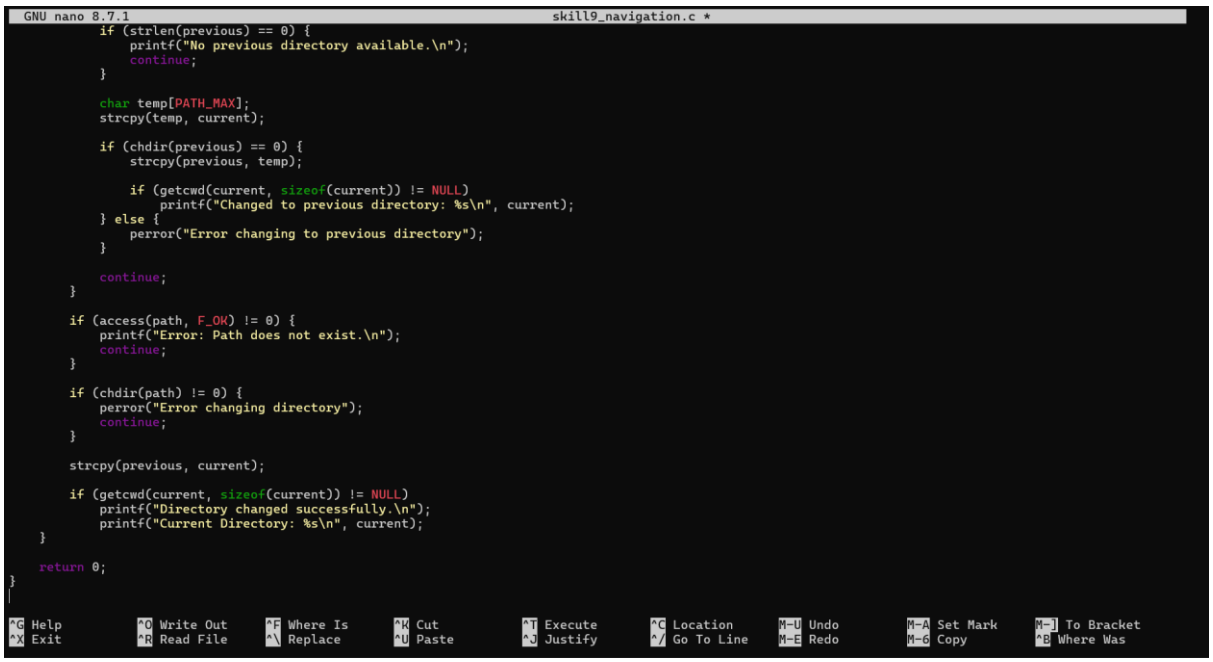
        path[strcspn(path, "\n")] = '\0';

        if (strcmp(path, "exit") == 0) {
            printf("Exiting navigation...\n");
            break;
        }

        if (strcmp(path, "pwd") == 0) {
            if (getcwd(current, sizeof(current)) != NULL)
                printf("Current Directory: %s\n", current);
            continue;
        }

        if (strcmp(path, "prev") == 0) {

^G Help          ^O Write Out     ^F Where Is      ^K Cut           ^T Execute       ^C Location      M-U Undo         M-A Set Mark     M-] To Bracket
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go To Line    M-E Redo         M-6 Copy         ^B Where Was
```



```
 GNU nano 8.7.1                                  skill9_navigation.c *
            if (strlen(previous) == 0) {
                printf("No previous directory available.\n");
                continue;
            }

            char temp[PATH_MAX];
            strcpy(temp, current);

            if (chdir(previous) == 0) {
                strcpy(previous, temp);

                if (getcwd(current, sizeof(current)) != NULL)
                    printf("Changed to previous directory: %s\n", current);
            } else {
                perror("Error changing to previous directory");
            }

            continue;
        }

        if (access(path, F_OK) != 0) {
            printf("Error: Path does not exist.\n");
            continue;
        }

        if (chdir(path) != 0) {
            perror("Error changing directory");
            continue;
        }

        strcpy(previous, current);

        if (getcwd(current, sizeof(current)) != NULL)
            printf("Directory changed successfully.\n");
            printf("Current Directory: %s\n", current);
    }

    return 0;
}

^G Help          ^O Write Out     ^F Where Is      ^K Cut           ^T Execute       ^C Location      M-U Undo         M-A Set Mark     M-] To Bracket
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go To Line    M-E Redo         M-6 Copy         ^B Where Was
```

Output:

```
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-09$ nano skill9_navigation.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-09$ gcc skill9_navigation.c -o skill9_navigation
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-09$ ./skill9_navigation
=== Directory Navigation ===
Current Directory: /home/uday/2520090089_OSSP/2520090089_Skill/Skill-09

Enter directory path (or 'prev' to go back, 'pwd' to display, 'exit' to quit): exit
Exiting navigation...
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-09$ ls
skill9_navigation  skill9_navigation.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-09$ ./skill9_navigation
=== Directory Navigation ===
Current Directory: /home/uday/2520090089_OSSP/2520090089_Skill/Skill-09

Enter directory path (or 'prev' to go back, 'pwd' to display, 'exit' to quit): pwd
Current Directory: /home/uday/2520090089_OSSP/2520090089_Skill/Skill-09

Enter directory path (or 'prev' to go back, 'pwd' to display, 'exit' to quit): skill9_navigation.c
Error changing directory: Not a directory

Enter directory path (or 'prev' to go back, 'pwd' to display, 'exit' to quit): prev
No previous directory available.

Enter directory path (or 'prev' to go back, 'pwd' to display, 'exit' to quit): ..
Directory changed successfully.
Current Directory: /home/uday/2520090089_OSSP/2520090089_Skill

Enter directory path (or 'prev' to go back, 'pwd' to display, 'exit' to quit): prev
Changed to previous directory: /home/uday/2520090089_OSSP/2520090089_Skill/Skill-09

Enter directory path (or 'prev' to go back, 'pwd' to display, 'exit' to quit): abc
Error: Path does not exist.

Enter directory path (or 'prev' to go back, 'pwd' to display, 'exit' to quit): pwd
Current Directory: /home/uday/2520090089_OSSP/2520090089_Skill/Skill-09

Enter directory path (or 'prev' to go back, 'pwd' to display, 'exit' to quit): exit
Exiting navigation...
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-09$
```

nano skill9_dispatch.c:

```
 GNU nano 8.7.1                                  skill9_dispatch.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>

#define MAX_INPUT 100

void cmd_pwd() {
    char cwd[PATH_MAX];

    if (getcwd(cwd, sizeof(cwd)) != NULL)
        printf("Current Directory: %s\n", cwd);
    else
        perror("pwd");
}

void cmd_cd(char *path) {
    if (path == NULL) {
        printf("cd: missing operand\n");
        return;
    }

    if (chdir(path) != 0)
        perror("cd");
    else
        printf("Directory changed successfully.\n");
}

void cmd_help() {
    printf("\nBuilt-in Commands:\n");
    printf("  pwd   - Display current directory\n");
    printf("  cd    - Change directory\n");
    printf("  help  - Display available commands\n");
    printf("  exit  - Exit the dispatcher\n");
}

void cmd_exit() {
    printf("Exiting command dispatcher...\n");
}

^G Help          ^O Write Out     ^F Where Is      ^K Cut           ^T Execute       ^C Location      M-U Undo         M-A Set Mark     M-] To Bracket
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go To Line    M-E Redo         M-6 Copy         ^B Where Was
```



Output:

```
=== Built-in Command Dispatcher ===

dispatch> help

Built-in Commands:
  pwd   - Display current directory
  cd    - Change directory
  help  - Display available commands
  exit  - Exit the dispatcher

dispatch> pwd
Current Directory: /home/uday/2520090089_OSSP/2520090089_Skill/Skill-09

dispatch> cd ..
Directory changed successfully.

dispatch> pwd
Current Directory: /home/uday/2520090089_OSSP/2520090089_Skill

dispatch> abc
Invalid command: abc
Type 'help' to see available commands.

dispatch> 2520090089_OSSP
Invalid command: 2520090089_OSSP
Type 'help' to see available commands.

dispatch> pwd
Current Directory: /home/uday/2520090089_OSSP/2520090089_Skill

dispatch> exit
Exiting command dispatcher...
```